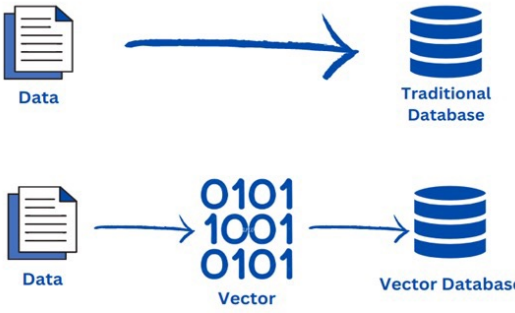
Vectors databases

- **Database** that **stores data** as **embeddings** (numerical vectors)
- Enables **semantic search** (based on **meaning**, not exact words)
- **Traditional DB** (SQL / NoSQL)
  - Works on: **exact match**, filters (WHERE, LIKE)



How Vector DB Works ?

1. Text → Embedding (Vectorization)

- Each **text** is converted into a **vector** using an **embedding model**

```
"Employees get 20 leaves" → [0.21, 0.78, 0.11, ...]
"Vacation policy" → [0.19, 0.80, 0.09, ...]
```

2. Store in Vector Database

- Each **record** contains:

```
{
  text: "Employees get 20 leaves",
  embedding: [0.21, 0.78, ...],
  metadata: { department: "HR" }
}
```

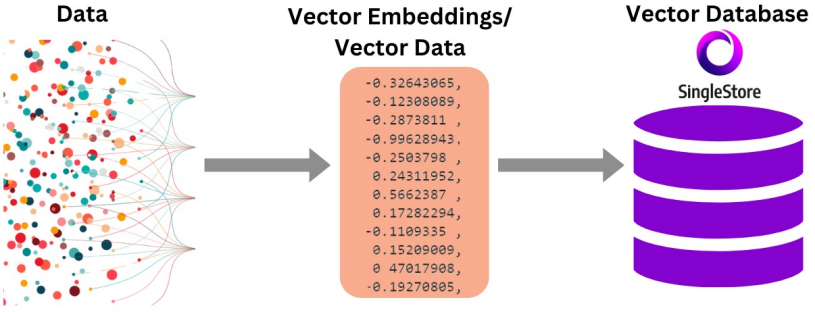
3. Query → Embedding

- User query: *“How many leaves do I get?”*
- Converted to:

```
[0.23, 0.81, ...]
```

4. Similarity Search

- Now **DB finds** vectors **closest** to **query vector**
- **ANN** (Approximate nearest neighbour) algorithm
- Common methods:
  - **Cosine similarity**
  - Dot product
  - Euclidean distance
- **Higher similarity** = **closer meaning**



5. Return Top Results

- Example result:

  "*Employees get 20 leaves*"

- This is then used by **LLM** (in RAG)

